

Topological Syntax in Killer Whale Communication

Executive Summary

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The Question

Do killer whales have grammar? And if so, what kind?

This study analyses approximately 40,000 killer whale vocalisations across four ecotypes from the DCLDE 2027 dataset (207,574 annotations, 23 hydrophone locations, 2005-2023). It introduces a novel analytical framework grounded in reaction-diffusion dynamics and tests a central hypothesis: that acoustic similarity between call types predicts their sequential order in natural vocal sequences.

The Core Finding: Topological Syntax

Calls that sound similar follow each other. This effect is large (Cohen’s $d = 1.34-1.57$), replicated across three genetically distinct populations (Southern Resident, Bigg’s Transient, Offshore), and robust to three independent controls for acoustic priming. At the highest similarity thresholds, sequentially adjacent calls are **136x more frequent** than chance.

This is fundamentally different from human language, where a word’s sound does not predict what comes next. In orca communication, it does. We term this **topological syntax**: grammar generated by the geometry of the signal space itself.

Key Results

Finding	Result	Significance
Topology predicts syntax	$d = 1.34-1.57$ across 3 ecotypes (8,570 calls)	Large effect, independently replicated
Sequential memory	MI half-life 45.4 calls (Bigg’s transients)	6-15x longer than any other non-human species
Markov order	>4 (bootstrap CI excludes zero)	Within the human language range (5-7)
Ecological stress detection	2019 prey crisis: bandwidth +328%, $p = 10^{-296}$	Population-specific, absent in control group
Linguistic universals	Zipf’s, brevity, Heaps’ laws all hold	SAR exponent (1.11) within human range

Validation

Before making novel claims, the framework was validated blind against Ford (1991): - **Correct** dialect ranking (J-L closest, K outlier) - **Correct** contact call identification (S01, from topology alone) - **Exact** repertoire recovery (shared and pod-specific call types)

The framework was not informed of the published answers.

Self-Correction

Two findings were retracted during internal review: - **Finding 29** (20ms inter-call interval): identified as a timestamp artifact in the Haro Strait dataset (0.2% in DCLDE vs 63% in Haro) - **Finding 43** (50-100ms fast response mode): same timestamp source, retracted for the same reason

These retractions, alongside two further honest nulls, are reported as evidence of the self-correcting methodology applied throughout.

The Orthogonality Argument

Human language and orca communication are not on the same complexity axis. They are **orthogonal solutions** to the problem of coordinating behaviour through sound:

Dimension	Human	Orca (Bigg's)
Vocabulary	~4.7 bits	0.9 bits
Sequential memory	3-8 words*	45.4 calls
Form-syntax coupling	~0	d = 1.57

*Unit mismatch noted: orca calls are analogous to utterances, not words. The non-human comparison (6-15x longer) uses equivalent units.

Human language excels at vocabulary, compression, and referential abstraction. Orca communication excels at sequential memory, topological coherence, and perceptual coordination. Neither is “more complex.” They solve different problems.

Conservation Application

The 2019 Chinook salmon crisis produced a measurable shift in SRKW communication across every dimension analysed, confirmed as population-specific by a natural experiment control (SAR, a healthy resident population, showed no shift). Passive acoustic monitoring using existing hydrophone infrastructure can detect population-level stress in real time, without visual surveys, without boats, without weather windows. For a population of ~75 individuals classified as Endangered, this has immediate operational value.

Methodology and Reproducibility

- 30 analysis scripts, each self-contained with fixed random seeds
- 74-test validation suite
- All data publicly available (DCLDE: NOAA DOI 10.25921/15ey-mh50; Ford-Osborne: Orcasound)
- Standard statistical methods throughout (k-means, MI, Mann-Whitney U, JSD, chi-squared, exponential regression)
- Pre-submission internal review identified and addressed four methodological vulnerabilities
- AI disclosure: scripts developed with Claude assistance; all scientific decisions are the author's

Full paper: 33 pages, 10 sections, 40+ references. Available on request.